

Chapter 3 supplementary material

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This document accompanies Chapter 3 of

Flora, D.B. (2026). *Statistical modelling methods: A guide for the behavioural and social sciences* (2nd ed.). Sage Publications.

Using Contrast-Code Variables

In the marital status example, perhaps a researcher explicitly planned a comparison of *single* individuals with all current or formerly married individuals – *married*, *divorced*, and *widowed* taken together as an aggregate. It is not possible to test this complex comparison using dummy variables; instead, it can be tested using **contrasts**.

The complex comparison for the marital status example implies the null hypothesis that the population mean *satis* score for *single* individuals (μ_1) equals the average of the population mean mean for the *married* (μ_2), *divorced* (μ_3), and *widowed* (μ_4) groups:

$$H_0: \mu_1 = \frac{\mu_2 + \mu_3 + \mu_4}{3},$$

which can be rearranged to

$$H_0: \mu_1 - \frac{\mu_2 + \mu_3 + \mu_4}{3} = 0,$$

or

$$H_0: 1\mu_1 - \frac{1}{3}\mu_2 - \frac{1}{3}\mu_3 - \frac{1}{3}\mu_4 = 0.$$

The latter expression of the complex null hypothesis establishes the weights for a linear combination of population means, that is, a **linear contrast**:

$$\psi_1 = 1\mu_1 - \frac{1}{3}\mu_2 - \frac{1}{3}\mu_3 - \frac{1}{3}\mu_4.$$

In general, a linear contrast for a set of K population means is any linear function of the form

$$\psi_k = c_1\mu_1 + c_2\mu_2 + \dots + c_K\mu_K = \sum_{k=1}^K c_k\mu_k$$

under the condition that the weights, or contrast codes, sum to zero, that is,

$$c_1 + c_2 + \dots + c_K = \sum_{k=1}^K c_k = 0.$$

Notice that the above contrast from the marital status example (ψ_1) meets the requirement that the contrast codes sum to zero ($1 - 1/3 - 1/3 - 1/3 = 0$). These coefficients for the population means in the linear contrast then become the observed values of a variable C_1 used in a regression of the outcome variable. That is, we assign $C_1 = 1$ for *single* individuals and $C_1 = -1/3$ for *married*, *divorced*, and *widowed* individuals.

But, similar to dummy coding, it is necessary to define $K - 1$ of these contrast-code variables so that the complete K -level predictor variable is represented in the multiple regression model. Most often, these contrasts are established so that they are *mutually orthogonal*, which is explained later. For any given application, there are many potential approaches to specifying contrast codes (especially with $K = 4$ or more categories). The marital status example presented herein provides an example of **Helmert contrasts**, in which the first contrast is a comparison of the first group or category to an average of the other categories, the second contrast compares the second category to the average of the subsequent categories (or just the last category if there are only three) ignoring the first category, and so on if there are four or more categories of the independent variable. Contrasts also may be specified in a more customized manner that does not

adhere to the pattern implied by Helmert contrasts (see Maxwell et al., 2018, for other approaches).

Thus, continuing the marital status example, we might specify a second Helmert contrast that compares the *satis* mean for currently *married* individuals with the aggregate mean of the *divorced* and *widowed* individuals, corresponding to the null hypothesis

$$H_0: \mu_2 = \frac{\mu_3 + \mu_4}{2}$$

or, equivalently,

$$H_0: 1\mu_2 - 0.5\mu_3 - 0.5\mu_4 = 0.$$

To see how this null hypothesis explicitly leads to a corresponding contrast, we can also incorporate the population mean of the *single* group:

$$H_0: 0\mu_1 + 1\mu_2 - 0.5\mu_3 - 0.5\mu_4 = 0.$$

The *single* group has a weight of zero because the null hypothesis is a comparison of *married* with *divorced* and *widowed*, ignoring the *single* group. The weights for the population means in the null hypothesis then give the contrast

$$\psi_2 = 0\mu_1 + 1\mu_2 - 0.5\mu_3 - 0.5\mu_4.$$

Now, the coefficients for the population means in ψ_2 become the observed values of a second contrast variable C_2 such that $C_2 = 0$ for *single* individuals, $C_2 = 1$ for *married* individuals, and $C_2 = -1/2$ for *divorced* and *widowed* individuals.

The final Helmert contrast for the marital status example compares the mean of the *divorced* individuals with the *widowed* mean. The corresponding null hypothesis is

$$H_0: \mu_3 - \mu_4 = 0.$$

Incorporating the population means of the *single* and married groups, the null hypothesis is:

$$H_0: 0\mu_1 + 0\mu_2 + 1\mu_3 - 1\mu_4 = 0,$$

and so the weights for the population means in the null hypothesis give the contrast

$$\psi_3 = 0\mu_1 + 0\mu_2 + 1\mu_3 - 1\mu_4.$$

The coefficients for the population means in ψ_3 provide the observed values for the contrast variable C_3 such that $C_3 = 0$ for *single* and *married* individuals, $C_3 = 1$ for *divorced* individuals, and $C_3 = -1$ for *widowed* individuals.

At this point, the contrasts ψ_1 , ψ_2 , and ψ_3 provide observed values for the $K - 1 = 3$ contrast-code variables (C_1 , C_2 , and C_3) needed for a regression of *satis* scores on the four-category independent variable, marital status, as summarized in the **contrast matrix** in Table S3.1. Note how the values in the columns of the contrast matrix correspond to the weights of the contrasts ψ_1 , ψ_2 , and ψ_3 ; it may also be helpful to compare Table S3.1 to Table 3.2 to see how the contrast codes compare with dummy codes for this same research example. Finally, C_1 , C_2 , and C_3 become the regressors in a multiple regression model of the *satis* outcome variable:

$$\hat{Y}_i = \beta_0 + \beta_1 C_{1i} + \beta_2 C_{2i} + \beta_3 C_{3i},$$

which again is almost always estimated using OLS.

Table S3.1

Contrast matrix showing assignment of contrast-code values to marital status categories

Marital status	Contrast variable values		
	C_1	C_2	C_3
<i>single</i>	1	0	0
<i>married</i>	-1/3	1	0
<i>divorced</i>	-1/3	-1/2	1
<i>widowed</i>	-1/3	-1/2	-1

Generalizing beyond the marital status example, an independent variable with K categories can be completely represented with a set of $K - 1$ contrast-code variables. The contrasts are constructed to allow planned comparisons among the means of the groups formed by the independent variable, including complex comparisons in which the aggregate of two or more groups is compared to either another group or another aggregate of groups. Each contrast ψ_k of the $K - 1$ linear contrasts is constructed so that its numerical codes sum to 0; as demonstrated for the marital status example, articulating the null hypothesis corresponding to a given contrast can guide its construction. After the contrast codes are used to form contrast variables (i.e., $C_1, C_2, \dots, C_{K-1}$), such that each case is assigned the values corresponding to their category, these $K - 1$ contrast variables then comprehensively represent the original, categorical independent variable in the regression model:

$$\hat{Y}_{ij} = \beta_0 + \beta_1 C_{1i} + \dots + \beta_{K-1} C_{(K-1)i}$$

or

$$Y_{ij} = \beta_0 + (\sum_{k=1}^{K-1} \beta_k C_{ki}) + \varepsilon_{ij}.$$

Although estimates of the partial slope parameters β_k lead to tests of the comparisons indicated by the contrasts, direct interpretation of their values is tricky, though, as demonstrated below.

When the contrast variable model presented above is fitted to the marital status data, the resulting R^2 and its F -test are *identical* to those obtained when the dummy-coding approach was used to model the same data (i.e., $R^2 = .14$ with $F(3, 32) = 1.71, p = .18$). In general, the representation of a K -category predictor variable using $K - 1$ contrast codes is equivalent to using $K - 1$ dummy codes in that the two approaches produce the same decomposition of outcome variable variance into model and error sources and the same predicted values (i.e., model-implied population means) for the K groups. Table S3.2 gives complete results for this new (but equivalent) model, showing that the predicted values obtained using contrast variables are the same as those obtained using dummy variables as reported in the main text. But the specific parameter estimates and the corresponding inferential conclusions are different from those obtained earlier with dummy variables, as described next.

Table S3.2

OLS regression results for marital status example with contrast variables

Regressor	$\hat{\beta}$	$SE(\hat{\beta})$	t	p	95% CI
(intercept)	80.38	1.09			(78.15, 82.60)
C_1	-1.49	1.89	0.79	.44	(-5.33, 2.35)
C_2	2.93	1.74	1.69	.10	(-0.61, 6.46)
C_3	2.04	1.59	1.28	.21	(-1.19, 5.26)

$$\hat{Y}_{married} = 80.38 - 1.49\left(-\frac{1}{3}\right) + 2.93(1) + 2.04(0) = 83.80$$

$$\hat{Y}_{divorced} = 80.38 - 1.49\left(-\frac{1}{3}\right) + 2.93\left(-\frac{1}{2}\right) + 2.04(1) = 81.44$$

$$\hat{Y}_{single} = 80.38 - 1.49(1) + 2.93(0) + 2.04(0) = 78.89$$

$$\hat{Y}_{widowed} = 80.38 - 1.49\left(-\frac{1}{3}\right) + 2.93\left(-\frac{1}{2}\right) + 2.04(-1) = 77.38$$
Note. $N = 36$.

The estimated intercept of the regression of *satis* scores on the contrast-code variables is $\hat{\beta}_0 = 80.38$. When an outcome variable is regressed on Helmert contrast-code variables, the intercept estimate equals to the mean of the individual group means. Thus, here the estimated intercept of 80.38 equals the average of $\bar{Y}_{married}$, $\bar{Y}_{divorced}$, $\bar{Y}_{single}$, and $\bar{Y}_{widowed}$. Next, the slope estimate for contrast variable C_1 is $\hat{\beta}_1 = -1.49$, which represents the complex comparison of the *single* group with the *married*, *divorced*, and *widowed* groups (i.e., ψ_1). Because this estimate is nonsignificant, $p = .44$, the null hypothesis for this contrast cannot be rejected. In other words, the single mean does not significantly differ from the average of the means of the *married*, *divorced*, and *widowed* groups. Likewise, the slope estimates $\hat{\beta}_2$ and $\hat{\beta}_3$ are also nonsignificant, indicating that the null hypotheses for the ψ_2 and ψ_3 contrasts also cannot be rejected based on the data.

Regarding the regression coefficients themselves, the estimate $\hat{\beta}_1 = -1.49$ equals the product of a constant (specifically, $c = 3/4$) and the difference between the sample mean *satis* score of the *single* group and the average of the sample means of the other three groups. That is, we have

$$\hat{\beta}_1 = c \left(\bar{Y}_{single} - \frac{\bar{Y}_{married} + \bar{Y}_{divorced} + \bar{Y}_{widowed}}{3} \right) = \frac{3}{4} \left(78.89 - \frac{83.80 + 81.44 + 77.38}{3} \right) = -1.49.$$

Likewise, the estimate $\hat{\beta}_2 = 2.93$ equals a constant (specifically, $c = 2/3$) times the difference between the sample mean *satis* score of the *married* group and the average of the divorced and widowed group means:

$$\hat{\beta}_2 = c \left(\bar{Y}_{married} - \frac{\bar{Y}_{divorced} + \bar{Y}_{widowed}}{2} \right) = \frac{2}{3} \left(83.80 - \frac{81.44 + 77.38}{2} \right) = 2.93.$$

Thus, although tests of the slope estimates for the two contrast code variables provide inferential conclusions for the contrast null hypotheses, the slope estimates themselves (and corresponding confidence intervals) are not directly interpretable when the contrasts are formed in the manner that led to the contrast matrix in Table S3.1.

Improved contrasts

We can instead obtain more directly interpretable regression coefficients using the new contrast codes given in Table S3.3 (for explanation of this issue, see Cohen et al., 2003, pp. 333-

334). In this revised contrast matrix, it is apparent that C_1 still represents a complex comparison of the *single* group with the aggregate of the other three groups, C_2 still represents a comparison of the *married* group with the *divorced* and *widowed* groups, and C_3 is a comparison of the *divorced* and *widowed* groups (these are still Helmert contrasts); additionally, each contrast still meets the requirement that the sum of its codes equals zero. But this revised contrast matrix was devised to satisfy another rule, namely that the difference between the value of the set of positive contrast codes and the value of the set of negative contrast codes should equal 1 for each contrast. In Table S3.3, that is the case because for C_1 we have $3/4 - (-1/4) = 1$, for C_2 we have $2/3 - (-1/3) = 1$, and for C_3 we have $1/2 - (-1/2) = 1$.

Table S3.3

Revised contrast matrix showing assignment of contrast-code values to marital status categories

Marital status	Contrast variable values		
	C_1	C_2	C_3
<i>Single</i>	3/4	0	0
<i>Married</i>	-1/4	2/3	0
<i>Divorced</i>	-1/4	-1/3	1/2
<i>Widowed</i>	-1/4	-1/3	-1/2

Table S3.4

OLS regression results for marital status example with improved contrast variables

Regressor	$\hat{\beta}$	$SE(\hat{\beta})$	t	p	95% CI
(intercept)	80.38	1.09			(78.15, 82.60)
C_1	-1.98	2.51	0.79	.44	(-7.11, 3.14)
C_2	4.39	2.60	1.69	.10	(-0.91, 9.69)
C_3	4.07	3.17	1.28	.21	(-2.39, 10.53)

Note. $N = 36$.

When cases are assigned values for the contrast variables based on this revised contrast matrix and the regression model is estimated again, different parameter estimates are obtained, as shown in Table S3.4, but R^2 remains .14 and readers can confirm for themselves that the predicted values, or model-implied means, remain the same. The intercept estimate is still $\hat{\beta}_0 = 80.38$, again equaling the average of the four group means. But the slope estimate for C_1 is now $\hat{\beta}_1 = -1.98$, which is exactly equal to the difference between the sample mean of the *single* group and the average of the sample means of the other three groups:

$$\hat{\beta}_1 = \left(\bar{Y}_{single} - \frac{\bar{Y}_{married} + \bar{Y}_{divorced} + \bar{Y}_{widowed}}{3} \right) = \left(78.89 - \frac{83.80 + 81.44 + 77.38}{3} \right) = -1.98.$$

Additionally, the slope estimate for C_3 is now $\hat{\beta}_2 = 4.39$, which is equals to the difference between the sample mean of the *married* group and the average of the divorced and widowed group means:

$$\hat{\beta}_2 = \left(\bar{Y}_{married} - \frac{\bar{Y}_{divorced} + \bar{Y}_{widowed}}{2} \right) = \left(83.80 - \frac{81.44 + 77.38}{2} \right) = 4.39.$$

Finally, $\hat{\beta}_3 = 4.07$ equals the difference between the *divorced* and *widowed* means.

Another advantage of using the current, revised contrast codes is that confidence intervals around the slope estimates are also directly interpretable in terms of the mean comparisons

specified by the contrast null hypotheses. Yet, the t -tests for the slope estimates are identical to those obtained with the first set of contrast codes, indicating that this alternative set of contrast-code variables leads to the same failed rejections of the null hypotheses behind contrasts ψ_1 , ψ_2 , and ψ_3 .

It is desirable for a given set of contrasts used to represent the categories of a categorical predictor to be **mutually orthogonal**, meaning that each individual contrast accounts for a separate, unique proportion of variance from among the overall proportion of outcome variable variance explained by the K -category predictor. Two contrasts are orthogonal if the products of the contrast codes within each category of the predictor sum to zero. In the marital status example, the contrasts in Table S3.3 are orthogonal to each other because the products of the values in each row of the contrast matrix sum to zero across the four categories of the independent variable: $(3/4*0*0) + (-1/4*2/3*0) + (-1/4*-1/3*1/2) + (-1/4*-1/3*-1/2) = 0$. When the sample sizes within each category of the independent variable are equal, the contrast-code variables C_k are uncorrelated with each other; with unbalanced sample sizes (as with the marital status example), the C_k will have non-zero correlations. Unbalanced sample size is usually presented as an important obstacle in traditional ANOVA analyses, but in the current multiple regression approach to ANOVA, unbalanced n across the levels of a categorical predictor is handled automatically through the partialling of regression coefficients to account for correlations among the contrast-code variables.